

Repeat Donor Classifier

Who do we call?

Gradient Boosting Classifier | 423,576 Donors | AUC = 0.796 | 85% Accuracy

NYC Campaign Finance Board | 2013–2021 Elections | 760,955 Transactions

WHAT PREDICTS WHETHER SOMEONE GIVES AGAIN

01 Size of First Donation

33.6%

return rate for \$500–\$1,000 first gifts

vs just 24% for under \$25 — the single biggest signal

02 Election Year of First Gift

32.6%

return rate for 2013 cohort

vs 24.1% for 2021 — earlier donors are far more loyal

03 Offices Donated To

Multi-office

signals broader civic engagement

Giving to multiple offices is a strong loyalty indicator

04 Borough

33.4%

Manhattan return rate

vs 22.5% for Bronx — geography shapes donor loyalty



HIGH Repeat Probability

Manhattan Executive Director who gave \$500+ in 2013 to multiple campaigns

- First gift \$500+ → highest return tier
- 2013 first gift → most loyal cohort (32.6%)
- Manhattan based → 33.4% borough return rate
- Multiple offices → proven cross-campaign engagement



LOW Repeat Probability

Bronx retiree who gave \$10 to a single candidate in 2021 for the first time

- First gift under \$25 → lowest return tier (24%)
- 2021 first gift → least loyal cohort (24.1%)
- Bronx based → 22.5% borough return rate
- Single candidate → no cross-campaign engagement

THE QUESTION — A DIFFERENT MODEL FOR A DIFFERENT PROBLEM

Our previous Ridge Regression predicted HOW MUCH a donor would give. This new classifier predicts WHETHER a donor will give again — a fundamentally different problem requiring different features.

Previous Model

Model Type

Regression

Question

How much will this donor give in total?

Top Predictor

Median gift amount dominates everything

This New Model

Model Type

Classification

Question

Will this donor donate again?

Top Predictor

First gift amount, year, office count

MODEL PERFORMANCE — GRADIENT BOOSTING CLASSIFIER

Trained on 338,860 donors, tested on 84,716. The model correctly identifies repeat donors and high-risk churners from first-touch data alone.

0.796

AUC Score
(vs 0.5 random baseline)

85%

Overall Accuracy
on held-out test set

27.4%

Base Rate — donors
who give again

100%

Precision on
Repeat Donors

Model Comparison

Logistic Regression

AUC: 0.768

Random Forest

AUC: 0.793

Gradient Boosting ✓

AUC: 0.796

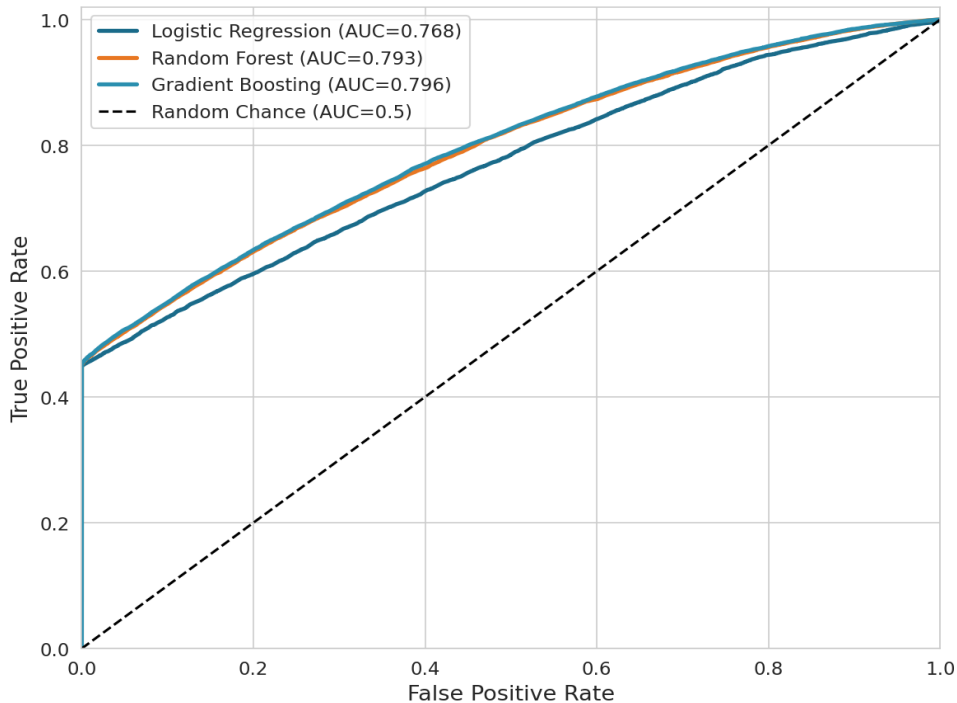
Insight

Gradient Boosting wins with AUC=0.796 — meaning if you randomly pick one repeat donor and one single-gift donor, the model ranks the repeat donor higher 79.6% of the time. Precision on repeat donors is 100% — when it says someone will give again, it is right every time.

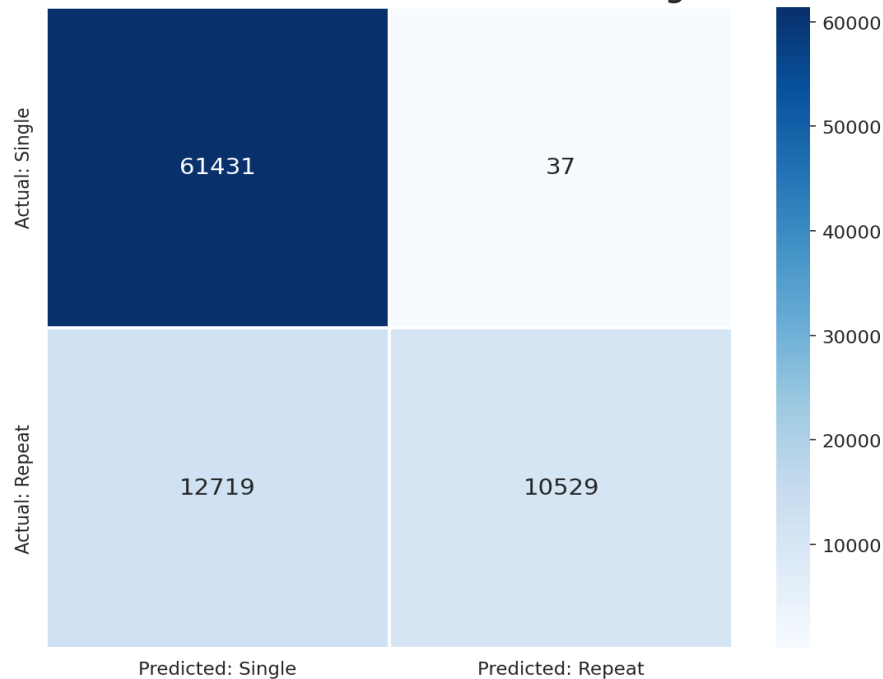
ROC CURVE & CONFUSION MATRIX — MODEL VALIDATION

The ROC curve shows strong separation above the random baseline. The confusion matrix reveals the model is conservative — highly precise when predicting repeat donors but captures about 45% of them (recall).

ROC Curves — All Models

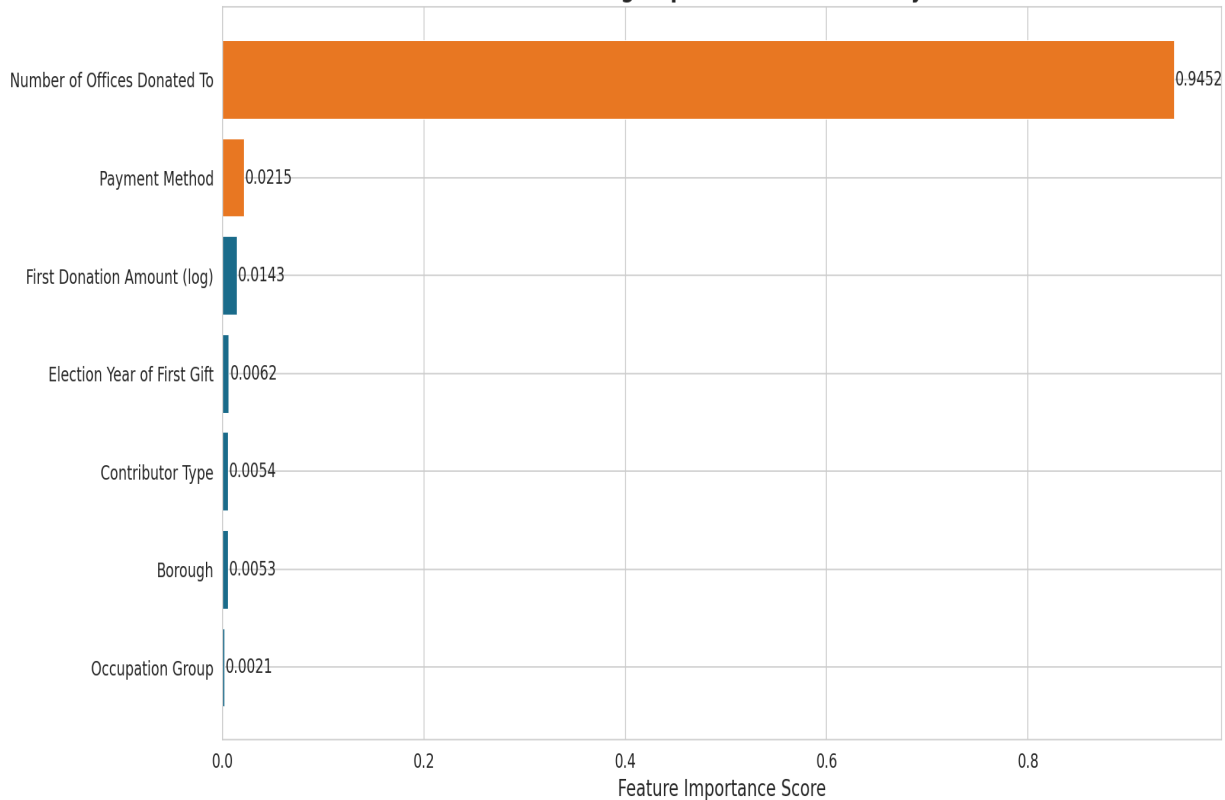


Confusion Matrix — Gradient Boosting



WHAT PREDICTS REPEAT GIVING — FEATURE IMPORTANCE

Feature Importance — Gradient Boosting
(Predicting Repeat Donor Probability)



Key Findings

First donation amount is #1 — bigger first gifts predict repeat giving

Election year matters — 2013 donors returned at 32.6% vs 24.1% for 2021

Number of offices donated to signals cross-campaign engagement

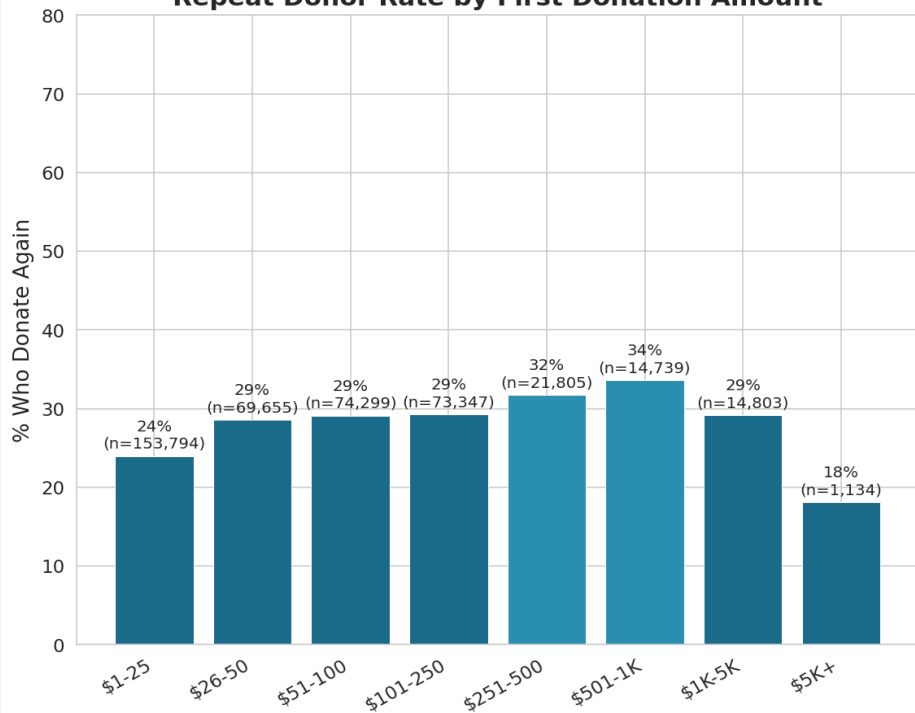
Borough, contributor type & payment method add smaller but real signals

Occupation group has the least marginal lift — consistent with value model

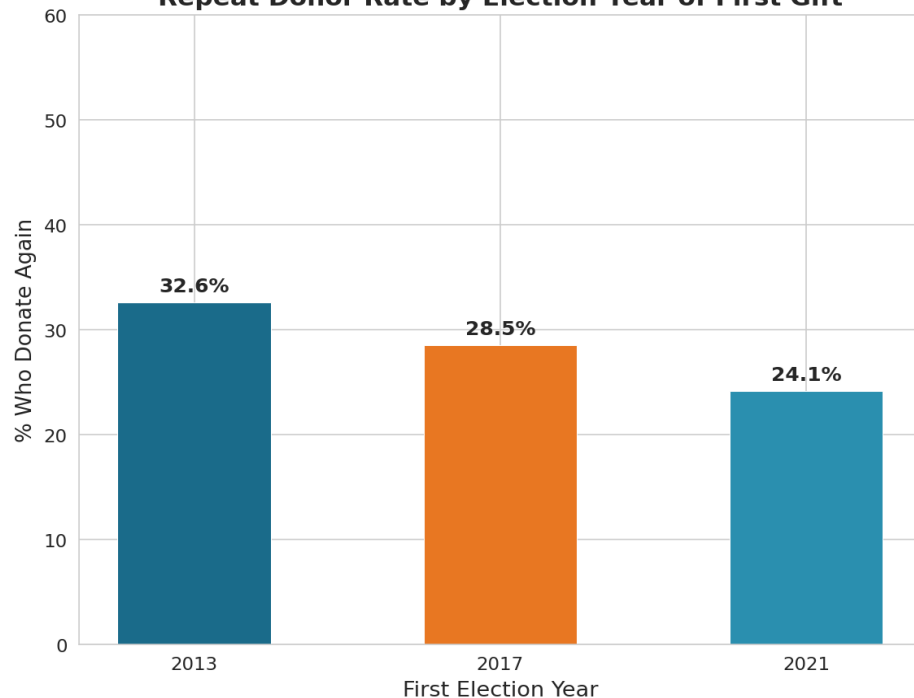
REPEAT RATES — FIRST DONATION AMOUNT & ELECTION YEAR

Donors who give \$500–\$1,000 on their first gift return at the highest rate (33.6%). 2013 cohort donors are the most loyal — nearly 1 in 3 came back across subsequent elections.

Repeat Donor Rate by First Donation Amount



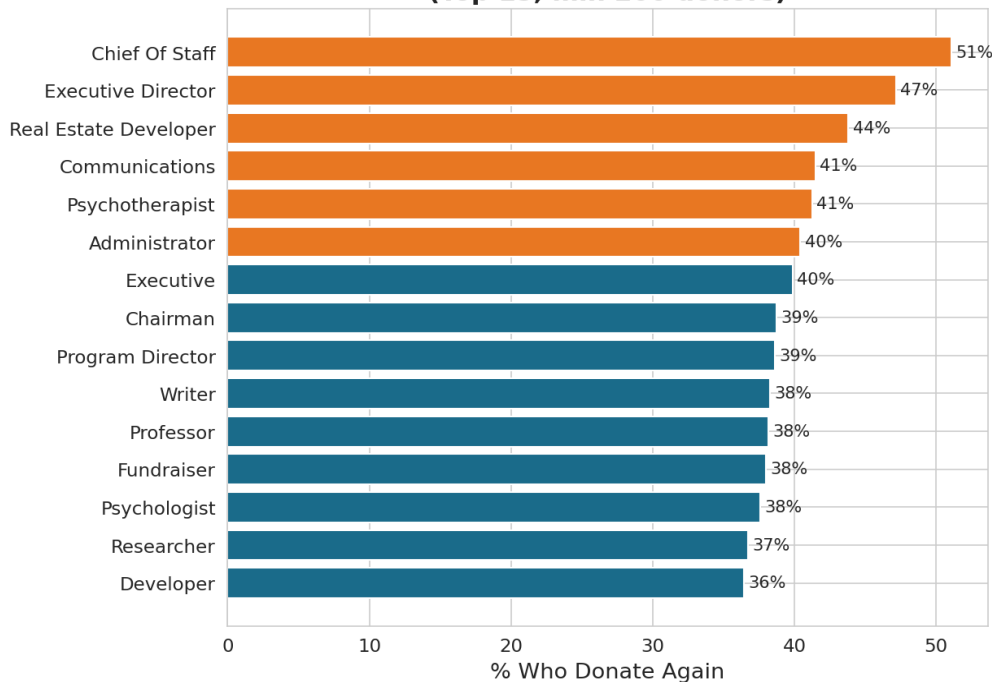
Repeat Donor Rate by Election Year of First Gift



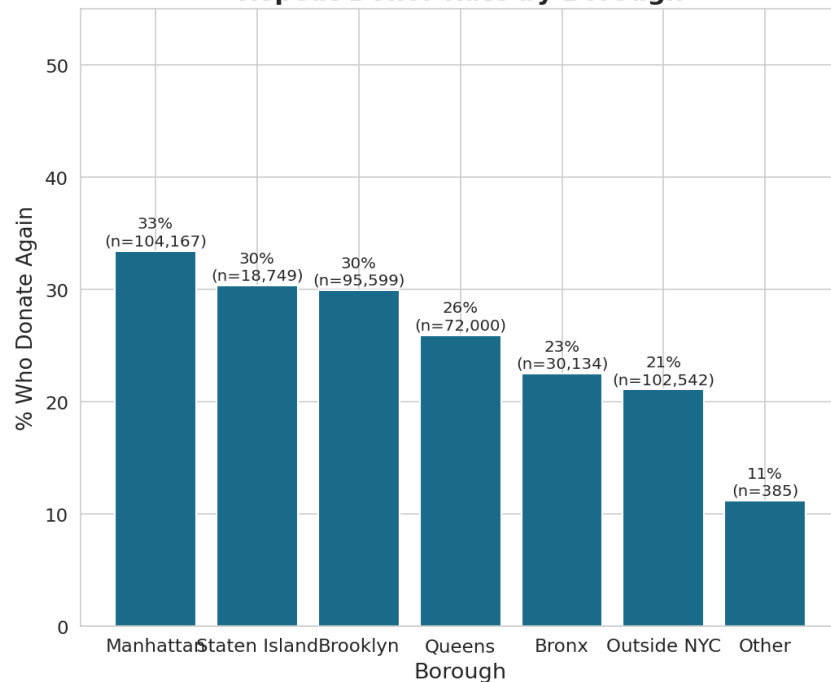
WHO COMES BACK — REPEAT RATE BY OCCUPATION & BOROUGH

Chief of Staff and Executive Director roles have the highest return rates (47–51%). Manhattan donors return at 33.4% — nearly 10 points higher than Bronx donors — suggesting geography and professional environment shape donor loyalty.

Repeat Donor Rate by Occupation
(Top 15, min 200 donors)

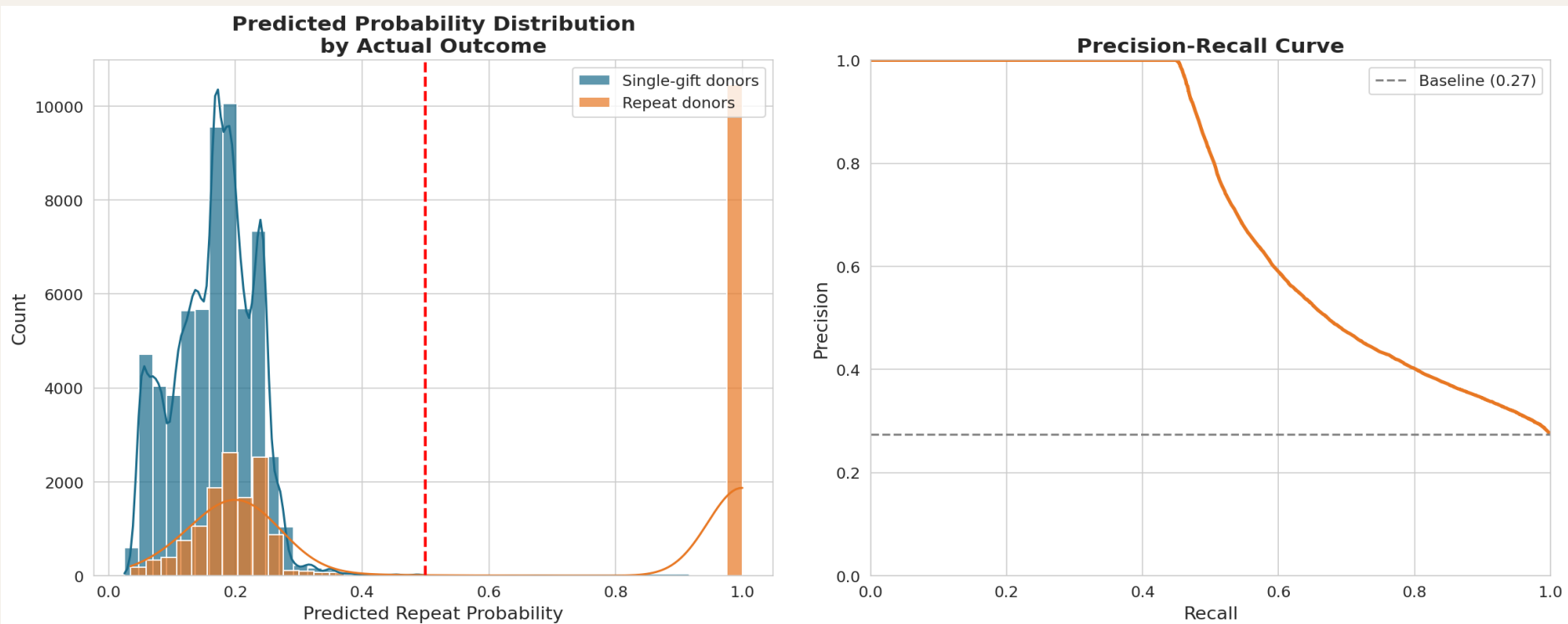


Repeat Donor Rate by Borough



PREDICTED PROBABILITY DISTRIBUTION & PRECISION-RECALL CURVE

The model assigns meaningfully different probability scores to repeat vs single-gift donors. The precision-recall curve shows strong performance above the 27.4% baseline — useful for tiered outreach campaigns.



HIGH vs LOW REPEAT PROBABILITY — DONOR PROFILES

Based on the model feature importance and repeat rates, two clear donor archetypes emerge for campaign targeting:



HIGH Repeat Probability

- ✓ First donation of \$250–\$1,000
- ✓ Donated first in 2013 election
- ✓ Manhattan or Staten Island based
- ✓ Executive Director, Chief of Staff, or Real Estate Developer
- ✓ Political Committee or Employer contributor type
- ✓ Donated to multiple offices or committees



LOW Repeat Probability

- First donation under \$25
- First donated in 2021 election
- Bronx or Outside NYC based
- Retired, Unemployed, or Student
- Individual contributor (IND)
- Gave to only one office or candidate

STRATEGIC RECOMMENDATIONS — PUTTING THE MODEL TO WORK

The classifier enables tiered, probability-driven donor outreach rather than blanket campaigns

01

Score Every First-Time Donor Immediately

Run the classifier within 24 hours of a first donation. High-score donors get premium follow-up — personalized outreach, phone calls, major-gift asks.

02

Prioritize 2013 Cohort for Re-engagement

Donors who first gave in 2013 return at 32.6% — the highest of any cohort. This is your most loyal base for any re-activation campaign.

03

Focus Manhattan Finance Professionals First

Manhattan donors return at 33.4%. Combined with occupation data, Executive Directors and Chiefs of Staff in Manhattan are the single highest-priority segment.

04

Combine Both Models for Full Donor Scoring

Use this classifier for who to contact, and the Ridge Regression for how much to ask. Together they give you a complete prioritization engine.